

## HYPOMAGNESEMIA DIAGNOSIS & MANAGEMENT

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### Magnesium Levels – Classification

| Severity        | Serum Magnesium (mmol/L) |
|-----------------|--------------------------|
| Normal          | 0.7 – 1.05               |
| Mild            | 0.5 – 0.69               |
| Moderate/Severe | < 0.5                    |

**⚠ Magnesium deficiency often coexists with other electrolyte imbalances. If potassium levels remain low despite supplementation, always evaluate magnesium status.**

### Clinical Features of Magnesium Deficiency

(Usually seen when  $Mg^{2+} < 0.5$  mmol/L)

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Muscles & Nerves    | Twitching, tremors, cramps, seizures, and muscle spasms                                       |
| Mental State        | Apathy, mood disturbances, hallucinations, confusion, and agitation                           |
| Heart & Circulation | Fast heart rate, high blood pressure, irregular rhythms, and increased sensitivity to digoxin |
| Lab Findings        | Often associated with low potassium, phosphate, calcium, and sodium levels                    |
|                     |                                                                                               |

### Identifying the Underlying Cause

Understanding the root cause is essential for both immediate treatment and preventing recurrence.

#### Medication-Related Causes

- Proton pump inhibitors (common)
- Diuretics (loop and thiazide types)
- Chemotherapy agents (e.g., Cisplatin)
- Aminoglycosides

- Amphotericin
- Immunosuppressants (e.g., Ciclosporin, Tacrolimus)
- Theophylline
- Beta-agonists (e.g., Salbutamol)

#### Gastrointestinal Causes

- Poor dietary intake (rare)
- Vomiting or chronic diarrhea
- Malabsorption (e.g., Coeliac disease)
- Laxative misuse
- Refeeding syndrome
- Alcohol dependency
- Bowel fistulas or short bowel syndrome
- Acute pancreatitis

#### Kidney-Related Causes

- Fluid overload
- Poorly controlled diabetes
- Alcohol-related kidney issues
- High calcium levels (e.g., primary hyperparathyroidism)
- Tubular dysfunction (e.g., post-transplant, recovery from acute kidney injury)
- Genetic syndromes (e.g., Bartter or Gitelman)

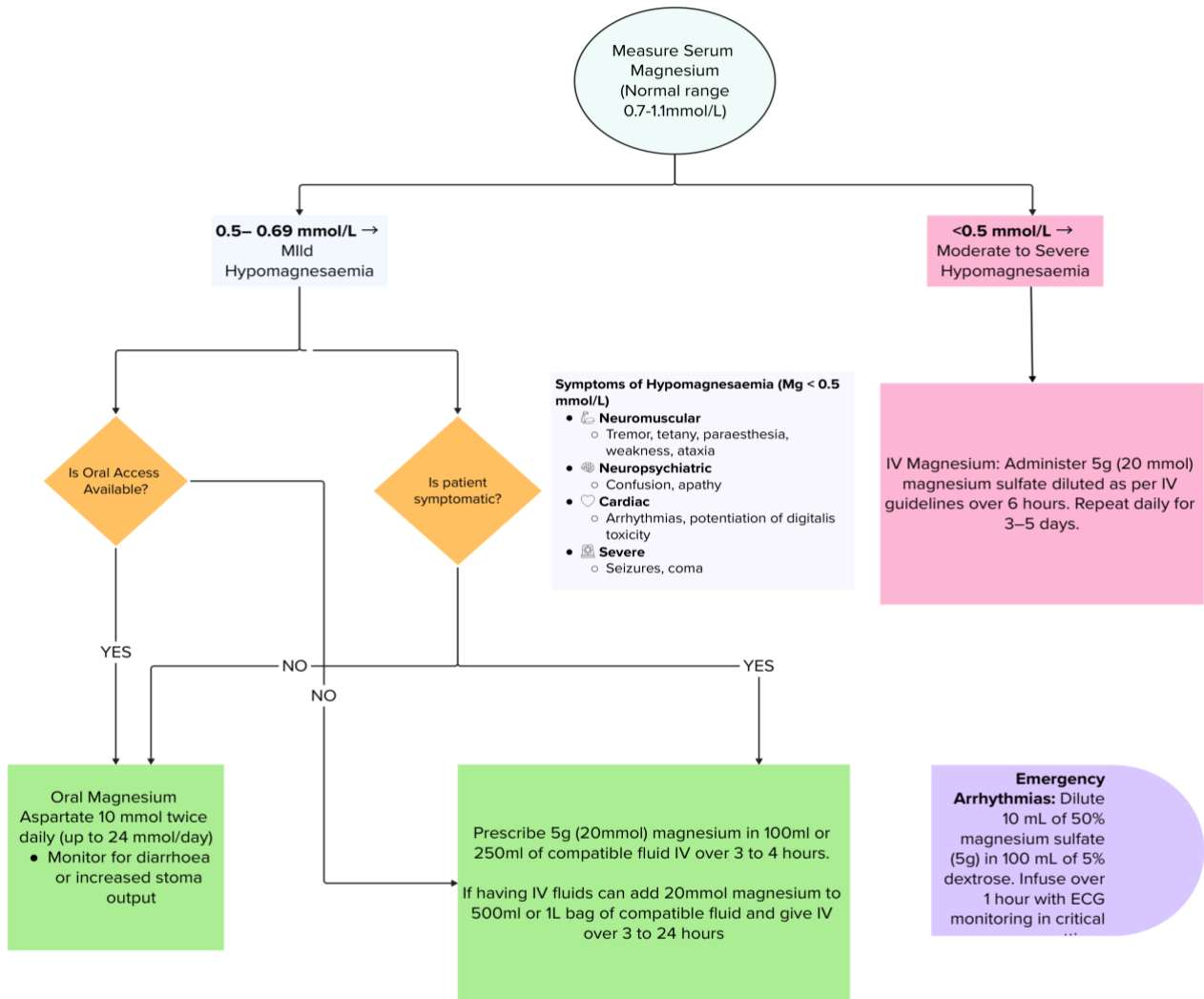
 **Note: Magnesium is primarily stored inside cells, so blood levels may appear normal even when the body is deficient.**

#### ECG Changes in Hypomagnesaemia

| ECG Change                        | Description                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------|
| Prolonged PR interval             | Delayed conduction between atria and ventricles                             |
| Widened QRS complex               | Prolonged ventricular depolarization                                        |
| Extended QT interval              | Prolonged ventricular repolarization                                        |
| Flattened T waves                 | Reduced amplitude of T waves                                                |
| Atrial or ventricular arrhythmias | Irregular heart rhythms                                                     |
| Torsades de pointes               | Specific type of ventricular tachycardia, especially in patients on digoxin |

# HYPOMAGNESEMIA MANAGEMENT GUIDELINES

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## Monitoring & Precautions

- **Vitals & ECG**
  - Monitor blood pressure, heart rate and respiratory rate
  - Obtain ECG if infusion rate exceeds 8 mmol/hour
- **Labs & Output**
  - Check serum magnesium daily
  - Monitor urine output closely (low output increases risk of Mg accumulation in renal impairment)
- **Contraindications**
  - Heart block or significant myocardial damage
  - Severe renal impairment
  - Myasthenia gravis
  - Hepatic failure
- **IV Precautions**
  - Avoid rapid bolus administration
  - Use a central line for concentrations > 20 mmol/100 mL

## Magnesium Supplementation in Renal Impairment and Deficiency

- **Risk with Renal Impairment:** Patients with severe renal insufficiency (eGFR < 30 mL/min/1.73 m<sup>2</sup>) are at increased risk of **severe hypermagnesemia** when treated with magnesium supplements.
  - Only replace magnesium if clinically necessary.
  - Use the **lowest effective dose**—reduce the usual dose by at least **50%**.
  - **Monitor serum magnesium levels** regularly to guide ongoing therapy.
  - **Do not use Magnaspartate®** in patients with **severe renal impairment** (eGFR < 30 mL/min).
  - Consider **IV magnesium** for patients with diarrhoea or **high-output stomas**.

## TREATMENT APPROACH?

### Step 1: Assess Severity

Is the magnesium level critically low?

Are there symptoms or ECG changes?

Are there life-threatening complications (e.g., arrhythmias, muscle weakness, unstable blood pressure)?

### Step 2: Choose Treatment Based on Severity

#### *Mild or Asymptomatic Cases*

Oral Magnesium: Use magnesium L-aspartate (Magnaspartate®). Each sachet provides 10 mmol of magnesium. Dose: 1–2 sachets daily depending on deficiency severity.

#### *Severe or Symptomatic Cases*

IV Magnesium: Administer 5g (20 mmol) magnesium sulfate diluted as per IV guidelines over 6 hours. Repeat daily for 3–5 days.

**Emergency Arrhythmias:** Dilute 10 mL of 50% magnesium sulfate (5g) in 100 mL of 5% dextrose. Infuse over 1 hour with ECG monitoring in critical care settings.

### Special Considerations

Kidney Impairment (eGFR < 30 mL/min/1.73 m<sup>2</sup>): Use magnesium cautiously. Reduce dose by at least 50% and monitor levels closely to avoid toxicity.

Cancer Patients: Refer to oncology-specific guidelines for magnesium management.

### Monitoring and Documentation

Record magnesium deficiency, its cause, and the treatment plan in the patient's medical notes.

In acute or symptomatic cases, cardiac monitoring is recommended.

Prefer IV therapy for patients with poor gut absorption or severe diarrhea.

Check magnesium levels daily, as blood levels may temporarily rise before stabilizing.

Monitor other electrolytes: calcium, sodium, potassium, and phosphate.

**Watch for signs of magnesium overdose, especially in elderly or renal-impaired patients:**

- Low blood pressure
- Slow heart rate
- Breathing difficulties
- Mental dullness
- Nausea/vomiting
- ECG abnormalities (e.g., heart block, asystole)